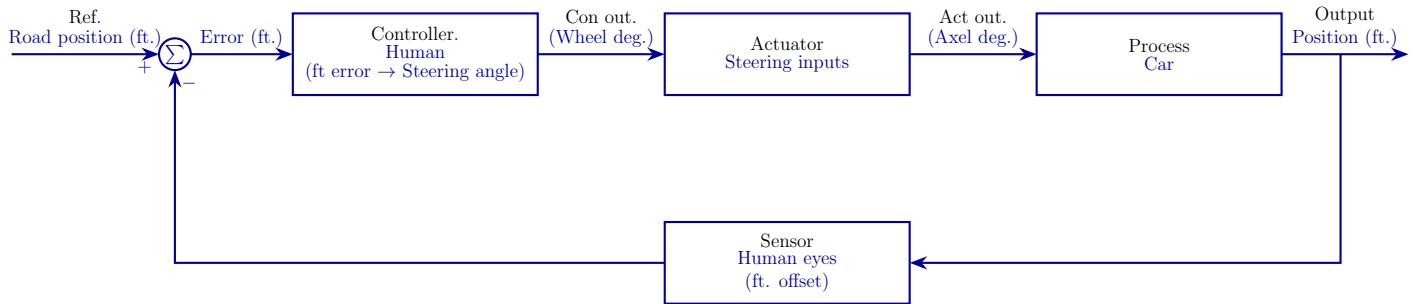


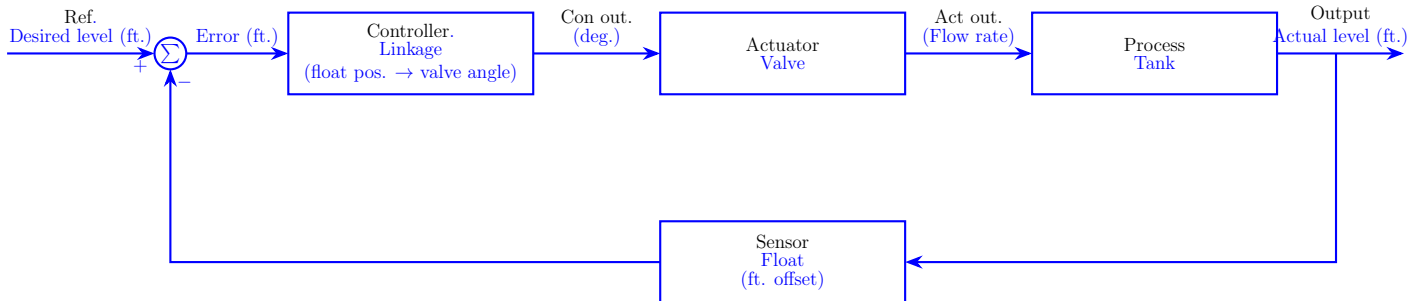
Problem 1 — 10 points

Draw a component block diagram for each of the following feedback control systems:

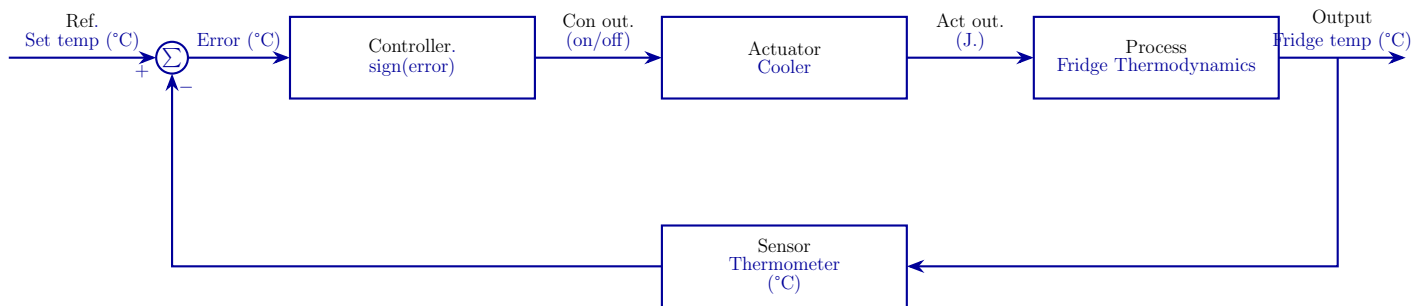
a. The manual steering system of an automobile



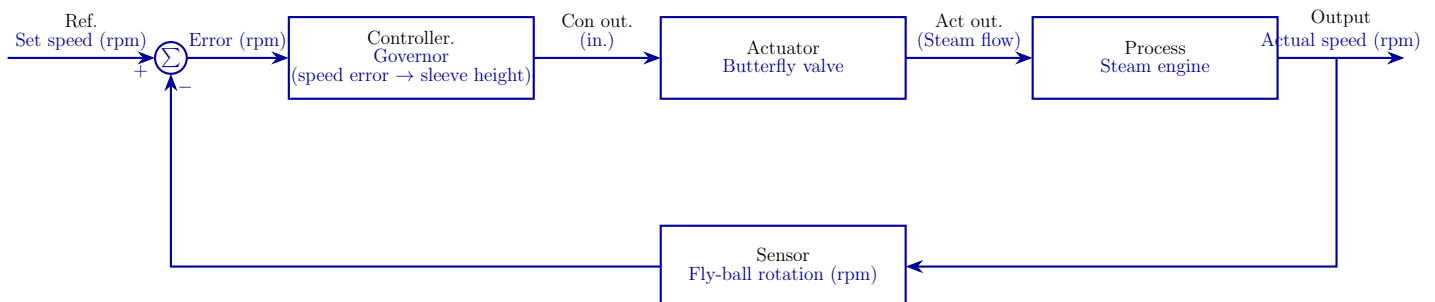
b. The water level controlled by a float and valve

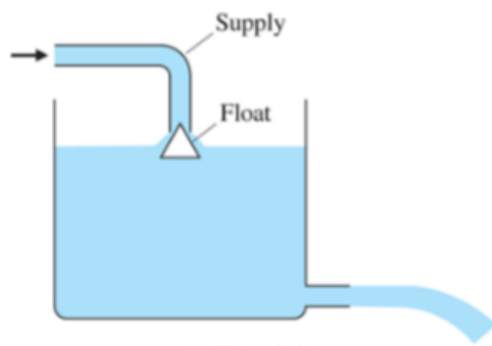


c. Temperature control in a refrigerator

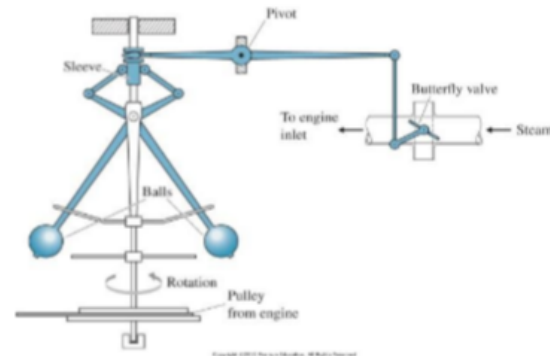


d. Watt's steam engine with fly-ball governor





Problem 1-b



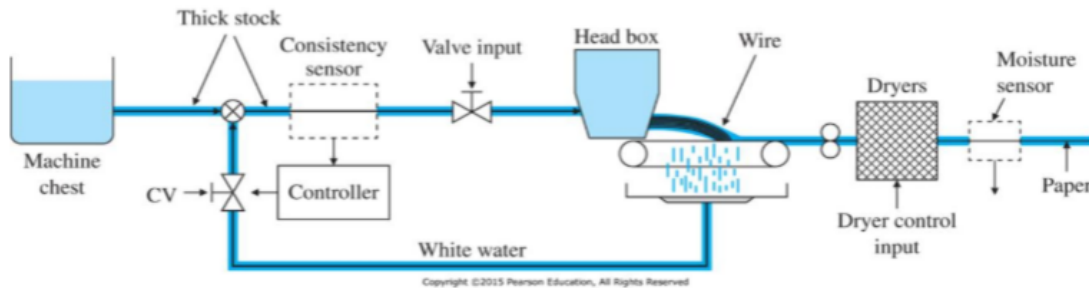
Problem 1-d

In each case indicate the location of the elements listed below and give the units associated with each signal:

- The process
- The process desired output signal
- The sensor
- The actuator & actuator output signal
- The controller & controller output signal
- The reference signal
- The error signal

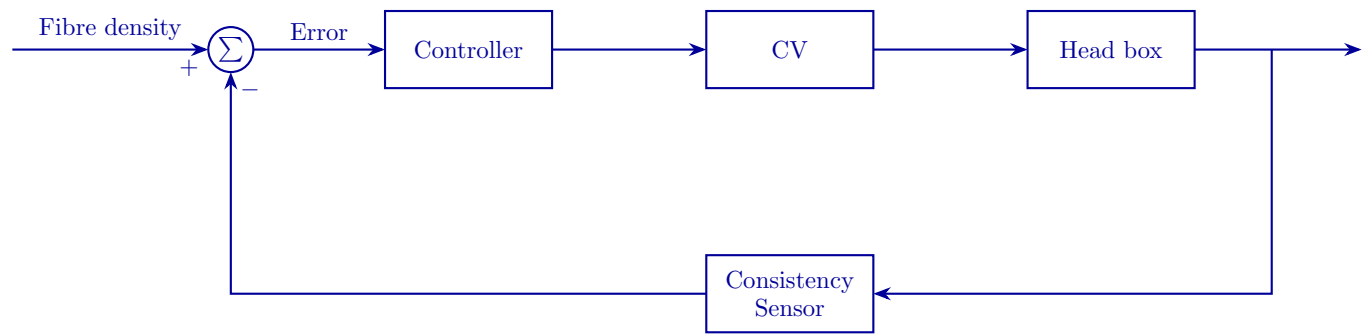
Problem 2 — 10 points

A machine for making paper is diagrammed below. There are two main parameters under feedback control: the density of fibers, as controlled by the consistency of the thick stock that flows from the headbox onto the wire, and the moisture content of the final product that comes out of the dryers. Stock from the machine chest is diluted by the white water returning from under the wire, as controlled by a control valve (CV).

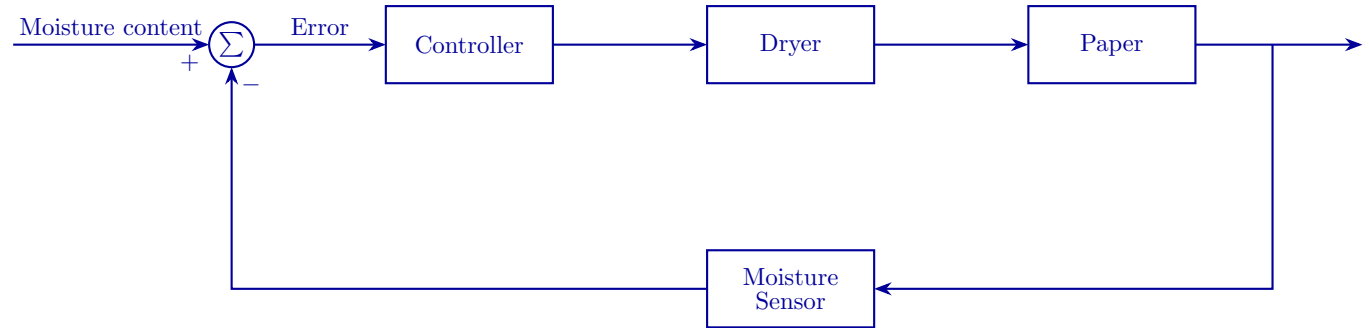


A meter supplies a reading of the consistency. At the “dry end” of the machine, there is a moisture sensor. Draw a block diagram and identify the components listed in Problem 1 (the process, etc.) for the following:

- Control of consistency



b. Control of moisture



Problem 3 — 10 points

Find the Laplace transform for the following time functions:

a. $f(t) = \sin(3t) + 2\cos(3t) + e^{-t}\sin(3t)$

$$\mathcal{L}[f] = \frac{3}{s^2 + 9} + \frac{6s}{s^2 + 9} + \frac{1}{(s + 1)^2 + 9}$$

b. $f(t) = t \sin t$

$$\mathcal{L}[f] = \frac{2s}{(s^2 + 1)^2}$$

c. $f(t) = 4 + 7t + t^2 + \delta(t)$

$$\mathcal{L}[f] = \frac{4}{s} + \frac{7}{s^2} + \frac{2}{s^3} + 1$$

Problem 4 — 10 points

Using the convolution integral, find the step response of the system whose impulse response is given below:

$$h(t) = \begin{cases} te^{-t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

Verify the results using the Laplace transform.

$$\begin{aligned}
y(t) &= \int_{-\infty}^{\infty} u(\tau)h(t-\tau)d\tau \\
&= \int_0^t (t-\tau)e^{-(t-\tau)}d\tau \\
&= e^{-t} \int_0^t (t-\tau)e^{\tau}d\tau \\
&= e^{-t} \left[(t-\tau)e^{\tau} \Big|_0^t + \int_0^t e^{\tau}d\tau \right] \\
&= e^{-t} [0-t+(e^t-1)] \\
&= e^{-t}(e^t-1-t) \\
&= \boxed{1-e^{-t}-te^{-t}}
\end{aligned}$$

Or, via Laplace:

$$\mathcal{L}[h(t)] = H(s) = \frac{1}{(s+1)^2}$$

so

$$Y(s) = H(s) \cdot U(s) = \frac{1}{(s+1)^2} \cdot \frac{1}{s} = \frac{1}{s(s+1)^2}$$

which is exactly

$$\mathcal{L}[1-e^{-t}-te^{-t}] = \frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2} = \frac{1}{s(s+1)^2}$$

Problem 5 — 10 points

Solve the following ODEs using the Laplace transform:

a. $\ddot{y}(t) + y(t) = t; \quad y(0) = 1; \quad \dot{y}(0) = -1$

$$\begin{aligned}
\mathcal{L}[\ddot{y} + y] &= \mathcal{L}[t] \\
s^2Y(s) - sy(0) - \dot{y}(0) + Y(s) &= \frac{1}{s^2} \\
(s^2 + 1)Y(s) - s + 1 &= \frac{1}{s^2}
\end{aligned}$$

$$\begin{aligned}
Y(s) &= \frac{1}{s^2(s^2+1)} + \frac{s-1}{s^2+1} \\
&= \frac{1}{s^2} - \frac{1}{s^2+1} + \frac{s}{s^2+1} - \frac{1}{s^2+1} \\
y(t) &= t - \sin t + \cos t - \sin t \\
&= \boxed{t + \cos t - 2\sin t}
\end{aligned}$$

b. $\ddot{y}(t) + 2\dot{y}(t) = e^t; \quad y(0) = 1; \quad \dot{y}(0) = 2$

$$\begin{aligned}
(s^2 Y(s) - s y(0) - \dot{y}(0)) + 2(s Y(s) - y(0)) &= \frac{1}{s-1} \\
(s^2 + 2s) Y(s) - (s+2)y(0) - \dot{y}(0) &= \frac{1}{s-1} \\
s(s+2) Y(s) - (s+2) - 2 &= \frac{1}{s-1} \\
Y(s) &= \frac{1}{(s-1)s(s+2)} + \frac{s+4}{s(s+2)} \\
&= -\frac{1}{2s} + \frac{1}{6(s+2)} + \frac{1}{3(s-1)} + \frac{2}{s} - \frac{1}{s+2} \\
y(t) &= -\frac{1}{2} + \frac{1}{6}e^{-2t} + \frac{1}{3}e^t + 2 - e^{-2t} \\
&= \boxed{-\frac{5}{6}e^{-2t} + \frac{1}{3}e^t + \frac{3}{2}}
\end{aligned}$$

Problem 6 — 10 points

Find the time function corresponding to each of the following Laplace transforms:

a. $F(s) = \frac{3s^2 + 6s + 6}{(s+1)(s^2 + 6s + 10)}$

$$\begin{aligned}
F(s) &= \frac{3s^2 + 6s + 6}{(s+1)(s^2 + 6s + 10)} \\
&= \frac{12s}{5(s^2 + 6s + 10)} + \frac{3}{5(s+1)} \\
&= \frac{12}{5} \frac{s}{(s+3)^2 + 1} + \frac{3}{5} \frac{1}{s+1} \\
&= \frac{12}{5} \left[\frac{s+3}{(s+3)^2 + 1} - \frac{3}{(s+3)^2 + 1} \right] + \frac{3}{5} \frac{1}{s+1} \\
f(t) &= \frac{12}{5} [e^{-3t} \cos t - 3e^{-3t} \sin t] + \frac{3}{5} e^{-t} \\
&= \boxed{\frac{12}{5} e^{-3t} (\cos t - 3 \sin t) + \frac{3}{5} e^{-t}}
\end{aligned}$$

b. $F(s) = \frac{e^{-s}}{s^2}$

$$\begin{aligned}
F(s) &= \frac{1}{s^2} e^{-s} \\
f(t) &= \boxed{(t-1)u(t-1)}
\end{aligned}$$

Note: You can use integration tables. Use the MATLAB Symbolic Algebra toolbox for verification.